

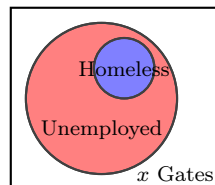
You must show work to receive credit. Unless indicated otherwise, each problem is worth equal value.

1. Construct a Venn diagram to determine the validity of the given argument.

- a. 1. All homeless people are unemployed.  
2. Bill Gates is not unemployed.

Therefore, Bill Gates is not homeless

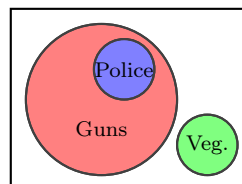
⇒ **Valid**



- b. 1. No vegetarian owns a gun.  
2. All policemen own guns.

Therefore, no policeman is a vegetarian.

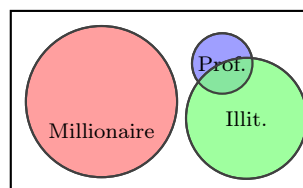
⇒ **Valid**



- c. 1. No professor is a millionaire.  
2. No millionaire is illiterate.

Therefore, no professor is illiterate.

⇒ **Invalid**



1. Which of the following are statements? Why or why not?

- a.  $3 + 7 = 9$

Yes, read “three plus seven is equal to nine”, and this is either true or false.

- b. Solve the equation  $3 + 7 = x$ .

No, it is a command. Commands are neither true nor false.

- c. Is  $\sqrt{2}$  an irrational number?

No, it is a question. Questions are neither true nor false.

2. Write a sentence that negates the statement.

- a. Some cats have three legs.

No cat has three legs.

- b. No cat has three legs.

Some cats have three legs.

- c. All of the cats have three legs.

Some cats do not have three legs.

- d. The cat is not blue.

The cat is blue.

3. Translate the sentence into symbolic form.

- a. It is a square and a rectangle.

$s \wedge r$

- b. If John is male then he is human.

$$m \rightarrow h$$

- c. All squares are rectangles.

$$s \rightarrow r$$

- d. All whole numbers are even or odd.

$$w \rightarrow (e \vee o)$$

- e. No square is a triangle.

$$s \rightarrow \sim t$$

- f. Being an orthodontist is sufficient for being a dentist.

$$o \rightarrow d$$

- g. Not being a monkey is necessary for being an ape.

$$a \rightarrow \sim m$$

4. Using the symbolic representations

$p$ : The car costs \$70,000.

$q$ : The car goes 140 mph.

$r$ : the car is red.

express the following compound statements in symbolic form.

- a. All red cars go 140 mph.

$$r \rightarrow q$$

- b. The car is red, goes 140 mph, and does not cost \$70,000.

$$r \wedge q \wedge \sim p$$

- c. If the car does not cost \$70,000, it does not go 140 mph.

$$\sim p \rightarrow \sim q$$

- d. The car is red, and it does not cost \$70,000 or go 140 mph.

$$r \wedge \sim (p \vee q)$$

- e. Being able to go 140 mph is sufficient for a car to cost \$70,000 or be red.

$$q \rightarrow (p \vee r)$$

- f. Not being red is necessary for a car to cost \$70,000 and not go 140 mph.

$$(p \wedge \sim q) \rightarrow \sim r$$